Abrakham A

Test Engineer, Active Safety (ADAS)
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SUMMARY

ADAS Active Safety validation engineer with 3.5+ years at TCS for General Motors. Wrote 2,000+ SIL test scripts across AEB, ACC, LKA, LCC, TSA, FCA, CIB and EBA, and built a YAML code generator that cut script-creation effort by 75%. Works in CAN/CAN-FD/CAN-TP, UDS, Python, CPPU unit testing and ISO 26262. Looking to move from SIL and unit-level validation into system-level ADAS testing.

SKILLS

- **ADAS and safety:** AEB, ACC, LKA, LCC, TSA, FCA, CIB, EBA, ISO 26262, ASIL
- **Test and automation:** Python, YAML code generation, SIL, regression testing, requirement traceability, CPPU unit test framework
- **HIL and simulation:** HIL test case preparation, dSPACE (exposure), Applied Intuition Object Sim, CarMaker, CarSim, RoadRunner
- **Networks and diagnostics:** CAN, CAN-FD, CAN-TP, UDS, LIN, Automotive Ethernet
- **Embedded software:** C, C++, Embedded C, MISRA-C, AUTOSAR (Classic), MATLAB/Simulink (basic)
- **Tools and process:** JIRA, RTC, PRTS, Confluence, Git, Eclipse, PDCA, LEAN, Agile

EXPERIENCE

Test Engineer, Active Safety Feb 2026 – Present
Tata Consultancy Services (Client: General Motors), Bengaluru

- Maintained 100% requirement traceability across ADAS active safety modules by deriving unit-level test cases from software and system requirements under ISO 26262-aligned methods, tracked in JIRA, RTC and PRTS.
- Raised test coverage from 55% to 100% and improved defect detection across active safety modules by integrating the CPPU unit test framework (Embedded C/C++, MISRA-C).
- Raised first-pass defect closure to 95% by running root cause and failure analysis with algorithm and embedded software teams and supplying logs and evidence from functional, regression and black-box tests.

SIL Validation Engineer, Active Safety Sep 2024 – Feb 2026
Tata Consultancy Services (Client: General Motors), Bengaluru

- Validated ADAS features (AEB, ACC, LKA, LCC, TSA, FCA, CIB, EBA) with 2,000+ test scripts covering functional and edge cases per ISO 26262 work products, written in Applied Intuition Object Sim and Eclipse.
- Cut manual test effort by 30% and regression run time by 80% by developing Python scripts for 15+ driving scenarios (highway merges, pedestrian crossings, emergency braking).
- Cut SIL script-creation effort by 75% by building a YAML-based code generator (Star of the Month, Q1 2025).
- Validated 8 UDS services by testing ECU and network behavior over CAN and CAN-TP in the SIL environment.
- Closed 12 defects across software releases by tracing failures to root cause with algorithm and embedded software teams and verifying the fixes.

Engineer, Active Safety Data Analysis Mar 2023 – Sep 2024
Tata Consultancy Services (Client: General Motors), Bengaluru

- Enabled the vehicle motion team to validate steering control on HIL by creating CAN/CAN-FD signal CGs and preparing HIL test cases.
- Analyzed 1,200+ defects across FCA, ACC, LCoD, ALC, CIB, AEB, EBA and Ped Alert by verifying ADAS behavior against real vehicle data, performing root cause and failure analysis, and reporting deviations from expected behavior to the client.
- Saved 4+ hours a week and gave the China, Korea and US teams one view of status by automating daily and weekly reports in Python.
- Standardized ADAS validation across three countries by training 18 engineers in China, Korea and the US.

EDUCATION

B.E., Electronics and Communication Engineering – Dhirajlal Gandhi College of Technology (Anna Univ.) 2022

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate (Oct 2025)
- Automotive Embedded Systems Training, SmartCliff Learning Solutions, Coimbatore (Jan 2024)
- Claude Associate Foundation Certificate, Anthropic; E1 Competency, Generative AI focus

ACHIEVEMENTS

- Star of the Month (Q1 2025) for the YAML-based code generator that cut script-creation effort by 75%.
- On the Spot Award for fast defect resolution during a critical AEB feature validation release.
- Slide Master Award for cross-functional reporting and stakeholder communication.